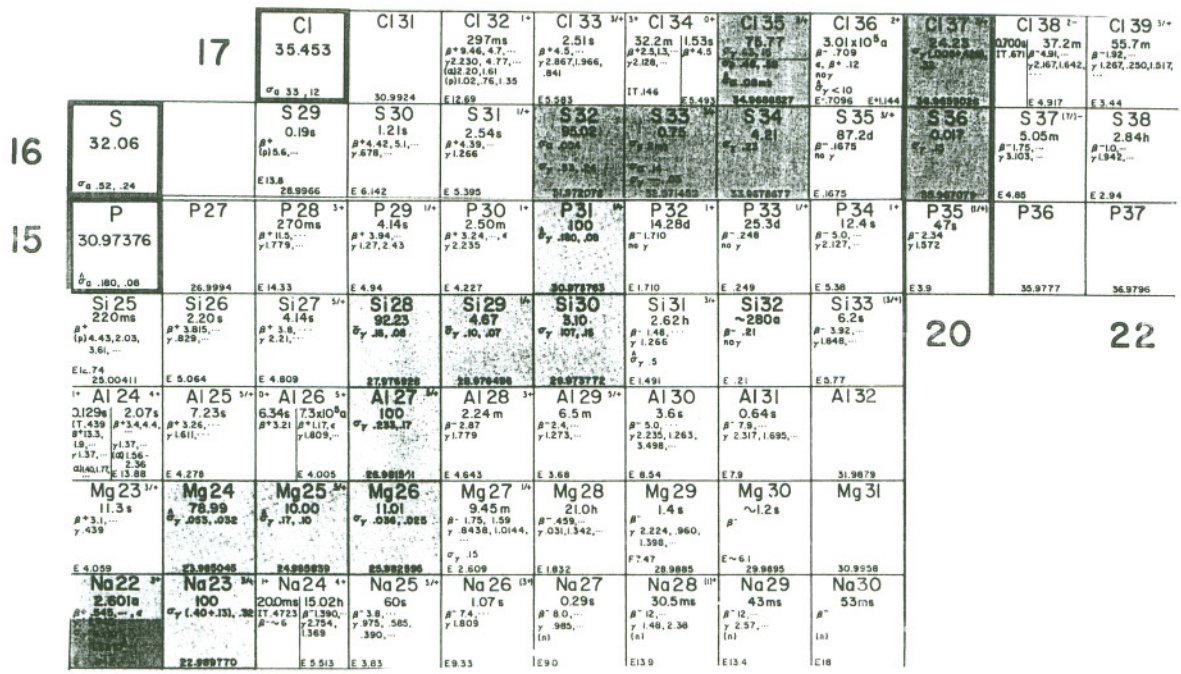
ANALYTICAL CUMULATIVE EXAM

November 13th, 1999

1. An important material in the semi-conductor industry is silicon doped with phosphorus. Using the excerpt from the Chart of the Nuclides below, design a procedure to produce this material involving neutron activation. Show that the amount of phosphorus required to produce is given by $P = \sigma \phi N_t (1 - e^{-\lambda t})$ where σ is the relevant cross section for Si, ϕ is the neutron flux (say 10^{12} cm^{-2} s^{-1}), N_t is the target atom density, λ is the relevant decay constant and t the irradiation time. How long is it necessary to irradiate to produce 1% silicon.

2. Describe a method by which the amount of phosphorus being produced could be experimentally determined <u>during the irradiation</u>.

3. What type of detector is best suited to the purpose in question 2?



Nuclear methods are commonly used to obtain high analytical precision for many kinds of geological and industrial materials. Above is an excerpt from the Chart of the Nuclides.

1. Sketch the mass parabolas for the isobaric series 235 and 238.

2. Using the parabolas, provide a qualitative explanation for the half-lives of the nuclides in the series. Also indicate the most likely form of decay displayed by each nuclide.

3. Besides decay along the mass parabolas, nuclides in these series display two other forms of decay. Discuss these? Use the Chart to provide a detailed discussion.

4. What, if any, are what are the common applications of these three decay modes.

5. How could you synthesize 1 g of ^{235}Np?

6. If the activity of ^{235}Np is currently 300 d.p.m., what activity would you expect after 7.5 year?